#4 Create an empty Bloom filter with the right size.
#5 Cycle through the list of contacts.
#6 For each contact, add it to the Bloom filter.

Once we set up our directory application, we have mainly two operations that we are interest in: checking if a contact is on the list, and adding a new contact to the directory.

For the former operation, shown in listing 4.2, we can check the Bloom filter, and if it says the contact has never been added, then we have our answer, the contact is not in the system; if, however, the Bloom filter returns `true`, then it might be a false positive, and so we need to contact the server to double check.

Listing 4.2 Checking an email

```
function checkContact(bloomFilter, server,
contact)                    #1
  if bloomFilter.contains(contact)
then                                  #2
    return
server.contains(contact)                                        #3
  else
    return false
                                                        #4
```

#1 Method `checkContact` verify if an email contact is stored in the application. It takes a Bloom filter, a server façade, and the contact to check. It returns true if contact is already in our contacts book.
#2 Check the Bloom filter for the contact passed to the method.